

P425/1
Pure Mathematics
Paper 1
July/August 2016
3 Hours

UGANDA ADVANCED CERTIFICATE OF EDUCATION
MOCK EXAMINATIONS
PURE MATHEMATICS
P425/1
3 HOURS

INSTRUCTIONS:

Attempt **all** questions in section **A** and any **five** in section **B**

Begin each answer on a fresh sheet of paper

No paper should be given for rough work

Mathematical tables with lists of formulae and squared paper are provided.

Silent non-programmable scientific calculators may be used.

State the degree of accuracy at the end of each question attempted using a calculator or tables, and indicate ***Cal*** for calculator or ***Tab*** for mathematical tables.

SECTION A: (40 MARKS)

1. Solve the equation $3^{2(1-x)} - 28(3^x) + 3 = 0$ (5 marks)
2. Evaluate $\int_0^{\pi/3} (1 + \cos 3x)^2 dx$ (5 marks)
3. Solve the equation, $4 \cos \theta - 2 \cos 2\theta - 3 = 0$ For $0^\circ \leq \theta \leq 360^\circ$ (5 marks)
4. Given that $y = \tan^{-1}\left(\frac{1-x}{1+x}\right)$, show that $\frac{dy}{dx} = \frac{-1}{1-x^2}$ (5 marks)
5. Prove by induction that $\sum_{r=1}^n 3^{r-1} = \frac{3^n - 1}{2}$ (5 marks)
6. The line L: $\frac{x+1}{2} = \frac{y-3}{3} = \frac{z+2}{-1}$ and the plane P: $x + y + z = 12$, intersect at a point A on the plane. Find the
 - (i) co-ordinates of A
 - (ii) angle between L and P
 (5 marks)
7. A triangle ABC has vertices at A, (-4, 10) B, (2,2), C, (5,8). D is the mid-point of AB. The line through D parallel to AC meets BC at E. Find the co-ordinates of the point E (5 marks)
8. Given that $y = \frac{2}{x^2}$ find $\frac{dy}{dx}$ from first principles (5 marks)

SECTION B: (60 MARKS)

9. Prove that in any triangle ABC $\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b} \cot\left(\frac{C}{2}\right)$
Hence solve the triangle in which $a = 9$, $b = 5.5$ and $C = 57^\circ$ (12 marks)
10. (a) Expand $(1-x)^{1/3}$ in ascending powers of x up to the term in x^3 . Use the expansion to evaluate $\sqrt[3]{998}$ (7 marks)
- (b) Determine the term independent of x in the expansion $\left[2x - \frac{1}{x^2}\right]^{12}$ (5 marks)
11. (a) Express $\frac{4x^2 - 2x + 3}{(x^2 + 1)(x - 2)}$ in partial fractions

Hence evaluate $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{4x^2 - 2x + 3}{(x^2 + 1)(x - 2)} dx$ (7 marks)

(b) Evaluate $\int_{-\pi/2}^{\pi/4} \frac{d\theta}{3\sin^2\theta + \cos^2\theta}$ (5 marks)

12. Show that for the line $yl + x + m = 0$ to be a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then $m^2 = a^2 + b^2l^2$. Hence determine the four common tangents to the two ellipses
 $E_1: 4x^2 + 14y^2 = 56$
 $E_2: 3x^2 + 23y^2 = 69$ (12 marks)

13. (a) Given that $Z = \frac{(1 - i)(2 + 3i)}{(3 + 4i)}$, Determine

- (i) Z in the form $x + iy$ and represent it on an Argand diagram
- (ii) the magnitude and argument of $(\bar{Z})^2$ (6 marks)

- (b) In an Argand diagram, W represents a complex number such that $\left| \frac{W - 1 - i}{W - 1 + i} \right|$

Show that W lies on a circle. Determine the area of this circle (Take $\pi = \frac{22}{7}$)

(6 marks)

14. A curve is given by $y = \frac{x(x - 3)}{(x^2 - 5x + 4)}$

- (i) State the asymptotes of the curve
- (ii) Show that the curve has no turning points and no restrictions on the value of y ; hence sketch the curve (12 marks)

15. A sample of a radioactive substance loses mass at a rate, which is proportional to the amount present. If M is the mass after time t , where t is in years ;

- (i) form a differential equation connecting M , t and constant k .
- (ii) if initially the mass of the substance is M_0 deduce that $M = M_0 e^{-kt}$
- (iii) given that its initial mass halved in 1600 years, determine the number of years it takes 15g of the substance to reduce to 13.6g? (12 marks)

16. (a) (i) A vector $\mathbf{r} = \begin{pmatrix} 1 \\ -\lambda \\ 4 \end{pmatrix}$ is perpendicular

to the line $\mathbf{r} = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -4 \\ 6 \end{pmatrix}$

Determine the value of λ .

(ii) Find the equation of a plane which contains a point with position vector $5\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ and perpendicular to the line in a(i) above. (8 marks)

(b) $\mathbf{p}$ and $\mathbf{q}$ represent the position vectors of the points P and Q respectively.

Show that the position vector of a point R which divides PQ in the ratio $\mathbf{m} : \mathbf{n}$ is represented by $\frac{n\mathbf{p} + m\mathbf{q}}{m + n}$ (4 marks)

END